Classical strip contraction and an attained threshold for the actual quadratic Xi flow

Project Stardust
A classical formalization companion

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Abstract

We reconstruct the classical coarse zero-strip bound for the quadratic heat deformation of the actual arithmetic Xi kernel. For the normalization $F_t(z) = \int_0^\infty K(u)e^{tu^2}\cos(zu)\,du$, every zero at $t \geq 0$ satisfies $(\operatorname{Im} z)^2 \leq \max(1-2t, 0)$. In particular all zeros are real for $t \geq 1/2$. The proof constructs real even divisor polynomials retaining the initial zeros and one arbitrarily small quadratic growth budget shared by all approximants. This budget justifies locally uniform imaginary-Gaussian heat convergence. A conjugate-root product identity contracts zero strips under symmetric imaginary shifts, and a finite-degree Euler limit gives the actual polynomial heat operator. A vertical modulus argument transfers the strip even across repeated roots and collisions. We include the exact Xi frequency normalization, positive origin value and original Gaussian integrability before passing to limits. These are classical de Bruijn mechanisms, already stated in broader form by Newman; no new strip bound, exact threshold, zero-time reality or Riemann-hypothesis result is claimed. Exact declarations and hypotheses are mapped to a frozen Lean library. The same actual flow has a finite attained all-real threshold; a complete four-point smoothing proof supplies the needed lower obstruction.

1 The arithmetic flow and the conclusion

For $v > 0$ write $\Theta_k(v) = \sum_{n\geq 1} n^k e^{-\pi n^2 v}$ and, for $u \geq 0$, set

$$K(u) = 2\pi^2 e^{9u}\Theta_4(e^{4u}) - 3\pi e^{5u}\Theta_2(e^{4u}). \tag{1}$$

The quadratic flow is the actual absolutely convergent integral

$$F_t(z) = \int_0^\infty K(u)e^{tu^2}\cos(zu)\,du, \qquad t \in \mathbb{R},\ z \in \mathbb{C}. \tag{2}$$

Time has the *growing* sign e^{tu^2}, so its equation is $\partial_t F_t = -\partial_z^2 F_t$. It is distinct from the positive-time Fourier multiplier e^{-tu^2} of ordinary real heat evolution. Let

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

with its entire continuation at the apparent exceptional points. Our frequency and scalar normalizations are

$$F_0(z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right). \tag{3}$$

Thus the elementary critical strip $0 \leq \operatorname{Re} s \leq 1$ corresponds to $|\operatorname{Im} z| \leq 1$, rather than width $1/2$.

Theorem 1.1. *For every real $t \geq 0$ and $z \in \mathbb{C}$,*

$$F_t(z) = 0 \quad \Longrightarrow \quad (\operatorname{Im} z)^2 \leq \max(1 - 2t, 0).$$

In particular F_t has only real zeros for every $t \geq 1/2$. Each F_t is an even real entire function and $F_t(0) > 0$.

The all-real conclusion includes multiplicities; it does not assert simplicity. The upper bound $1/2$ is coarse and classical. No assertion about the exact infimum of all-real times, or about zeros at time zero, is made. The fixed mathematical baseline is

`bab688bb6140cd6316c80680095a914ea0d346db`.

Newman's Theorem 1, printed page 246 [2], already states the broader classical strip contraction for even rapidly decaying Fourier kernels, crediting de Bruijn [1]. Our purpose is to expose how the actual arithmetic and approximation hypotheses in the verified library close that argument. This is a formalization exposition, not a new analytic theorem.

2 Integrability, positivity and the exact Xi normalization

The theta series and their derivatives converge uniformly on compact subsets of $v > 0$. For example any polynomial factor n^k is dominated by a constant times $e^{\pi n^2 a/2}$ on $v \geq a > 0$, leaving a summable Gaussian tail. Termwise differentiation consequently gives $\Theta_k' = -\pi\Theta_{k+2}$ there. Expanding (1) yields

$$K(u) = \sum_{n \geq 1} \pi n^2 e^{5u}(2\pi n^2 e^{4u} - 3)e^{-\pi n^2 e^{4u}} > 0 \qquad (u \geq 0),$$

since $2\pi > 3$. Moreover, for $v \geq 1$,

$$\Theta_k(v) \leq \Theta_k(1/2)e^{-\pi v/2}.$$

Indeed $n^2 v \geq n^2/2 + v/2$ for $n \geq 1$. Thus for a finite constant C,

$$|K(u)| \leq Ce^{9u - \pi e^{4u}/2} \quad (u \geq 0). \tag{4}$$

Every fixed nonnegative integer power of u, multiplied by $e^{Tu^a + Bu}$ with a fixed nonnegative integer a and real T, B, is integrable against this bound. This follows because e^{4u} dominates every fixed power of u; beyond a finite cutoff the full integrand is bounded by a constant times e^{-u}. On the remaining compact interval continuity supplies integrability.

For z in any fixed complex disk, $|\cos(zu)| \leq e^{|z|u}$, and each spatial derivative adds only a power of u. The same bound with a larger disk is a common integrable majorant. Differentiation under the integral therefore proves entireness, all spatial derivatives and the stated time equation. Evenness is literal; conjugation gives $F_t(\overline{z}) = \overline{F_t(z)}$. At zero the integrand is strictly positive, so $F_t(0) > 0$. In particular none of the functions to which the zero-limit argument will be applied is identically zero.

For completeness the normalization can be recovered without treating (3) as a new assumption about the kernel. Put

$$\psi(u) = e^u \Theta_0(e^{4u}).$$

Direct differentiation gives $K = (\psi'' - \psi)/8$. Jacobi's classical theta identity implies

$$\psi(u) - \psi(-u) = \frac{e^{-u} - e^u}{2}, \qquad \psi'(0) = -\frac{1}{2}.$$

The derivative follows by differentiating the first identity at zero. The functions ψ, ψ' decay at positive infinity with the same superexponential mechanism as (4). For real x the derivative of $\psi'(u)\cos(xu)+x\psi(u)\sin(xu)$ is $(\psi''(u)+x^2\psi(u))\cos(xu)$. Integrating and evaluating both boundary terms gives

$$F_0(x) = \frac{1}{16} - \frac{1+x^2}{8}\int_0^\infty \psi(u)\cos(xu)\,du. \tag{5}$$

The standard theta Mellin representation of the pole-removed completed zeta, split at one and transformed by the same Jacobi identity, is

$$\Lambda_0(s) = \int_1^\infty \Theta_0(v)\bigl(v^{s/2} + v^{(1-s)/2}\bigr)\,\frac{dv}{v}.$$

Here Λ_0 is the entire part after removing the two simple pole terms, so $\xi(s) = (s(s-1)\Lambda_0(s)+1)/2$. The integral converges for all s by exponential theta decay, and is the usual analytic continuation of its convergent Mellin representation. At $s = (1+ix)/2$, substitution $v = e^{4u}$ gives $\Lambda_0(s) = 8\int_0^\infty \psi(u)\cos(xu)\,du$. Since $s(s-1) = -(1+x^2)/4$, this identity and (5) prove (3) on the real axis. Both sides are entire, so the identity theorem proves it on the entire complex plane. The foundational inputs here are the classical Jacobi theta identity and Mellin representation; the displayed derivatives, substitutions and boundary evaluations are the actual normalization calculation in the source library.

The initial strip and a subquadratic bound

The Euler product gives $\zeta(s) \neq 0$ for $\operatorname{Re} s > 1$. Equation (3), the nonvanishing gamma factor and the functional equation therefore exclude zeros with $\operatorname{Im} z < -1$ or $\operatorname{Im} z > 1$. Apparent exceptional points in the completed formula do not introduce zeros: on the imaginary axis the actual integral is $\int K(u)\cosh(yu)du > 0$. Thus

$$F_0(z) = 0 \quad\Longrightarrow\quad |\operatorname{Im} z| \le 1. \tag{6}$$

No improved arithmetic zero-free half-plane is imported.

For fixed t, there is $C_t > 0$ such that

$$|F_t(z)| \le C_t e^{|z|^{3/2}} \qquad (z \in \mathbb{C}). \tag{7}$$

Here is an elementary bound matching the source exponent. If $R \ge 0$ and $u \ge 0$, then $Ru \le R^{3/2}+u^3$: for $u \le \sqrt{R}$ use $Ru \le R^{3/2}$; otherwise use $Ru \le u^3$. Therefore

$$|F_t(z)| \le e^{|z|^{3/2}} \int_0^\infty |K(u)| e^{tu^2+u^3}\,du,$$

and the integral is finite by (4). This is a genuine subquadratic entire growth bound, not only a real-axis estimate.

3 Real divisor polynomials with one common Gaussian budget

We use the classical Hadamard factorization theorem under its explicit finite-order hypotheses. An even entire f with $f(0) \neq 0$ and $|f(z)| \le Ce^{|z|^{3/2}}$ has the paired factorization

$$f(z) = f(0)\prod_{\rho\in\mathcal{Z}_+}(1 - z^2/\rho^2)^{m_\rho}. \tag{8}$$

Here $\mathcal{Z}_+$ contains one representative of each nonzero opposite zero pair, and m_ρ is the actual multiplicity. Products are ordered by expanding disks. Hadamard's genus-one elementary factors for

opposite zeros multiply to $1 - z^2/\rho^2$; the residual exponential has degree at most one, and evenness makes its linear coefficient zero. This is the classical entire-function input, not an asserted Lean theorem in the shorter notation above. McMullen's primary notes state the required Hadamard theorem and opposite-pair simplification [3].

The inverse squares are absolutely summable. One way to see the exact hypothesis is Jensen's formula: $f(0) \neq 0$ and the growth bound give $n_f(R) \leq C'(1 + R^{3/2})$ by applying Jensen at $2R$. On dyadic annuli,

$$\sum_{2^j \leq |\rho| < 2^{j+1}} \frac{m_\rho}{|\rho|^2} \leq \frac{n_f(2^{j+1})}{2^{2j}} = O(2^{-j/2}),$$

so summing the tail proves the assertion. There are finitely many zeros in bounded disks and none in a sufficiently small disk around zero. These facts justify uniform convergence of the paired product on compact sets and all finite products used below.

Apply this to $f = F_0$. Choose radii $R_n \to \infty$ avoiding zeros, and let

$$P_n(z) = F_0(0) \prod_{\substack{\rho \in \mathcal{Z}_+ \\ |\rho| < R_n}} (1 - z^2/\rho^2)^{m_\rho}. \tag{9}$$

Complete disks preserve all conjugate and opposite zeros, so P_n is a real even polynomial. It is nonzero because $P_n(0) = F_0(0) > 0$. Its zeros, with their multiplicities, are exactly the zeros of F_0 in that disk, and all satisfy (6). Also $P_n \to F_0$ locally uniformly. The source makes a definite expanding-radius choice $2(n+1) < R_n < 2(n+1) + 1$; no monotonicity of that particular choice is needed. Radii can always be chosen away from the discrete zero set.

Lemma 3.1. *For every $\epsilon > 0$ there is $C_\epsilon > 0$, independent of n, with*

$$|P_n(z)| \leq C_\epsilon e^{\epsilon |z|^2} \quad (n \geq 0,\ z \in \mathbb{C}).$$

Proof. Choose a finite initial set of paired zeros so the sum of $m_\rho/|\rho|^2$ outside it is at most $\epsilon/2$. For any finite subset of that tail,

$$\prod |1 - z^2/\rho^2|^{m_\rho} \leq \exp\left(|z|^2 \sum \frac{m_\rho}{|\rho|^2}\right) \leq e^{\epsilon |z|^2/2}.$$

For each of the finitely many initial factors, its polynomial growth is bounded by a constant times an arbitrarily small Gaussian. More explicitly, $1 + ar^2 \leq (1 + a/\delta) e^{\delta r^2}$ for $a \geq 0$, $\delta > 0$. Choose δ so the sum of the allocated multiplicity budgets is at most $\epsilon/2$. This bounds every subset of the finite initial product by one constant times $e^{\epsilon |z|^2/2}$, enlarging the constant to at least one for omitted factors. Combine head and tail and include $|F_0(0)|$. The same bound handles the finitely many early approximants and establishes a single constant for all n. □

The source proves this common budget by finite divisor jets and an absolute annular tail estimate in the initial strip. The Jensen/Hadamard presentation above is a labelled classical human reorganization of the same actual divisor-polynomial conclusion. It does not assume that F_0 has real or simple zeros.

4 The Gaussian heat limit is in its actual domain

Let $Y_s \sim N(0, 2s)$ for fixed $s \geq 0$. The actual flow obeys

$$\mathbb{E} F_t(z + iY_s) = F_{t+s}(z), \qquad \mathbb{E} F_t(z + Y_s) = F_{t-s}(z). \tag{10}$$

For a complex exponential, $\mathbb{E}e^{qY_s} = e^{sq^2}$. In the half-line Fourier representation this turns $e^{iu(z+iY_s)}$ into e^{iuz+su^2}. Every interchange is absolutely integrable: integrating the absolute value first in Y_s is bounded by a constant times $|K(u)|e^{(t+s)u^2+|\operatorname{Im} z|u}$, integrable by (4). The opposite-frequency term is handled identically. For real shifts the absolute-value bound is even smaller, and the Gaussian characteristic function yields e^{-su^2}. This proves both identities, including their time signs. At $s = 0$ the law is a point mass.

Proposition 4.1. *For every fixed $s \geq 0$,*

$$\mathbb{E}P_n(z + iY_s) \longrightarrow F_s(z)$$

locally uniformly in $z \in \mathbb{C}$. Each expectation is absolutely integrable. For each fixed $z, c \in \mathbb{C}$, the same approximants also converge to F_0 in the shifted Gaussian L^1 norm: $\mathbb{E}|P_n(z+cY_s) - F_0(z+cY_s)| \to 0$.

Proof. Assume $s > 0$; $s = 0$ is immediate from local uniform convergence. For a compact disk $|z| \leq R$, Lemma 3.1 and $|z + cy|^2 \leq 2R^2 + 2|c|^2y^2$ give

$$|P_n(z + cy)| \leq C_\epsilon e^{2\epsilon R^2} e^{2\epsilon|c|^2y^2}.$$

Choose $\epsilon > 0$ with $2\epsilon|c|^2 < 1/(4s)$. This is integrable under $N(0, 2s)$, uniformly in n and in that disk. The limit F_0 inherits the same bound. Dominated convergence proves the stated L^1 limit at each fixed z. To obtain local uniform integral convergence, first restrict to $|y| \leq A$. The affine images of $|z| \leq R$ form a compact set, so the integrand differences tend to zero uniformly there. Outside $|y| \leq A$, the displayed common Gaussian majorant bounds the integral tail uniformly in n, z, and that tail tends to zero as $A \to \infty$. Finally use (10) at $t = 0$, $c = i$. □

For polynomials define the finite operator

$$H_sP = e^{-s\partial_z^2}P := \sum_{j=0}^{\lfloor \deg P/2 \rfloor} \frac{(-s)^j}{j!} P^{(2j)}. \tag{11}$$

Taylor's finite formula and $\mathbb{E}Y_s^{2j} = (2j)!s^j/j!$ give exactly $H_sP(z) = \mathbb{E}P(z + iY_s)$. Thus the locally uniform limit in Proposition 4.1 is the actual finite polynomial operator limit, with no infinite-dimensional backward-heat series assumed convergent on an unspecified domain.

The common budget cannot be omitted. For $Y \sim N(0, 1)$ the polynomials $Q_n(z) = 1 + z^{2n}/(2n-1)!!$ tend to one on every compact complex set, but $\mathbb{E}Q_n(Y) = 2$ for every n. The compact convergence follows from the ratio $R^2/(2n+1) \to 0$ of successive nonconstant bounds on $|z| \leq R$; the expectation identity is the exact even Gaussian moment. Local uniform convergence by itself does not license this Gaussian averaging. This is a classical diagnostic, not an alternative Xi approximant.

5 Conjugate-root products contract strips under imaginary shifts

For a real nonzero polynomial P, let

$$C_hP(z) = \frac{P(z + ih) + P(z - ih)}{2}, \qquad h > 0.$$

Its leading coefficient and degree are unchanged, so it is nonzero and has real coefficients. Suppose all zeros of P obey $(\operatorname{Im} a)^2 \leq b^2$.

Lemma 5.1. *Every zero of C_hP satisfies*

$$(\operatorname{Im} z)^2 \le \max(b^2 - h^2, 0).$$

Proof. If P is constant the assertion is vacuous. Otherwise list its complex roots with multiplicity in a nonempty multiset L, invariant under conjugation. For $z = x + iy$, direct polynomial expansion gives

$$\begin{aligned} &|z + ih - a|^2|z + ih - \overline{a}|^2 - |z - ih - a|^2|z - ih - \overline{a}|^2 \\ &\quad = 8hy((x - \operatorname{Re} a)^2 + y^2 + h^2 - (\operatorname{Im} a)^2). \end{aligned} \tag{12}$$

If $y > 0$ and $y^2 + h^2 > b^2$, this difference is strictly positive for every root, including real roots and repeated roots. Multiplying the strict inequalities over L, all factors on the larger side are positive. Conjugation invariance makes each paired product a square of the unpaired product. Taking nonnegative square roots and restoring the nonzero leading coefficient gives

$$|P(z - ih)| < |P(z + ih)|.$$

The two terms cannot sum to zero. For a lower-half-plane candidate use conjugation of the real polynomial C_hP. Any nonreal zero violating the proposed bound has $y^2 > 0$ and $y^2 + h^2 > b^2$, so both half-planes are excluded. Real zeros already satisfy the nonnegative bound. □

Iterating the same shift n times therefore gives

$$C_h^n P(z) = 0 \quad \Longrightarrow \quad (\operatorname{Im} z)^2 \le \max(b^2 - nh^2, 0). \tag{13}$$

The budget identity is $\max(\max(B, 0) - q, 0) = \max(B - q, 0)$ for $q \ge 0$, proved by separating $B \ge 0$ and $B < 0$. Thus induction remains valid after the strip has collapsed. No root is followed through a collision, and no simple-root or distinct-root assumption is imposed at any stage.

6 The finite-degree shift limit and closure of zero strips

Fix $s > 0$ and a degree bound d. On the finite-dimensional space of polynomials of degree at most d, differentiation D is a bounded nilpotent operator. Taylor's formula gives the exact finite identity

$$C_h = \sum_{j=0}^{\lfloor d/2 \rfloor} \frac{(-1)^j h^{2j}}{(2j)!} D^{2j}.$$

Set $h_n = \sqrt{2s/n}$ for $n \ge 1$, and $A = -sD^2$. For any fixed coefficient-space norm,

$$C_{h_n} = I + A/n + E_n, \qquad \|E_n\| \le C/n^2.$$

There are only finitely many terms; the estimate includes $d < 4$, when the remainder is zero. Also $\|C_{h_n}\|, \|I + A/n\| \le 1 + C'/n$ after enlarging the constants. The difference-of-powers telescoping identity therefore bounds

$$\|C_{h_n}^n - (I + A/n)^n\| \le n\frac{C}{n^2}(1 + C'/n)^{n-1} \le \frac{Ce^{C'}}{n} \longrightarrow 0.$$

Finally A is nilpotent, so the binomial expansion of $(I + A/n)^n$ has a fixed finite number of terms. For each fixed j, $\binom{n}{j}/n^j \to 1/j!$. Hence

$$C_{h_n}^n P \longrightarrow \sum_j \frac{A^j P}{j!} = H_s P \tag{14}$$

in coefficients and locally uniformly on the complex plane. This finite-dimensional proof is a human reorganization of the formal Hasse-derivative/factorial-jet Euler limit, not a claim that the actual Xi function belongs to that bounded factorial coefficient domain.

We will use the following precise limit fact twice.

Lemma 6.1. *Let nonzero polynomials Q_n have all zeros in $|\operatorname{Im} z| \le b$, with $b \ge 0$. If $Q_n(z) \to f(z)$ pointwise on $\mathbb{C}$, and f is an entire function not identically zero, then all zeros of f lie in the same strip.*

Proof. For a factor $z - a$ with $\operatorname{Im} a \le b$, and real $b \le y_1 \le y_2$, its modulus at $x + iy_1$ is at most its modulus at $x + iy_2$, since $(y_1 - \operatorname{Im} a)^2 \le (y_2 - \operatorname{Im} a)^2$. Factorization into all roots with multiplicity gives

$$|Q_n(x + iy_1)| \le |Q_n(x + iy_2)|.$$

Pass to the pointwise limit. If $f(x + iy_2) = 0$ for $y_2 > b$, then $f(x + iy) = 0$ for every $y \in [b, y_2]$. The identity theorem makes f identically zero, a contradiction. Applying the same argument to $Q_n(-z)$ and $f(-z)$ excludes zeros below $-b$. □

This argument uses the given entireness and nonzero hypothesis of the limit. Pointwise limits in general need not be entire, and the zero function cannot be admitted. It avoids assuming convergence of individual zeros or a continuous ordering of roots.

Proposition 6.2. *For every nonzero real polynomial P, every $s \ge 0$, and every bound $(\operatorname{Im} a)^2 \le b^2$ on its initial zeros, the zeros of $H_s P$ obey $(\operatorname{Im} z)^2 \le \max(b^2 - 2s, 0)$.*

Proof. At $s = 0$ the operator is the identity. At $s > 0$, the scaled shifts in (14) all have this same strip by (13), since $nh_n^2 = 2s$. Their locally uniform limit $H_s P$ is nonzero: (11) preserves the leading coefficient. Apply Lemma 6.1 with $\sqrt{\max(b^2 - 2s, 0)}$ and square the resulting nonnegative bounds. □

7 Transfer to the actual Xi flow and exact scope

Proof of Theorem 1.1. The actual polynomials P_n of (9) are nonzero real even polynomials, and (6) bounds all their initial zeros by width one. Proposition 6.2 gives all zeros of $H_t P_n$ the common width $\sqrt{\max(1 - 2t, 0)}$. Proposition 4.1 identifies their locally uniform limit with the actual F_t, after establishing the necessary Gaussian integrability and common tail bound. That limit is entire and nonzero by Section 2. Lemma 6.1 transfers the strip, proving the claimed squared inequality. For $t \ge 1/2$ its right side is zero, forcing $\operatorname{Im} z = 0$. The symmetry and positive-origin claims were already proved directly from the kernel. □

The proof uses two different limiting operations. First, for each fixed polynomial, increasingly many small symmetric shifts give its finite heat operator, with degree fixed during that limit. Second, the genuine divisor-polynomial sequence has increasing degree and converges after Gaussian averaging because of the shared budget. No degree-uniform Euler estimate is needed, and no interchange of these two limits is asserted. This is the useful domain distinction in the formalization.

The coordinate conversion is likewise part of the conclusion. Newman's classical definition uses $s = 1/2 + iz$ and a damping parameter in e^{-bu^2} [2]. Here the Xi argument is $z/2$, the initial strip has width one, and forward t multiplies the arithmetic kernel by e^{+tu^2}. Numerical thresholds cannot be transferred between these conventions without rescaling both frequency and time. We use only the bound proved in the displayed normalization.

This manuscript does not prove reality at $t = 0$, an exact numerical value of the quadratic threshold, a sharper arithmetic zero-free region, or any relation between a higher-even-order threshold and RH. It also proves neither simplicity of all positive-time zeros nor an unrestricted backward-heat operator on an entire-function space. Local Hermite splitting and simple-root motion have their own verified hypotheses and are not needed here. These limits are mathematical scope statements, not unresolved steps in the proof of Theorem 1.1.

8 A finite attained threshold for the actual quadratic flow

Theorem 8.1. *There is a finite real number $\Lambda \le 1/2$ such that*

$$\{t \in \mathbb{R} : \textit{every zero of } F_t \textit{ is real}\} = [\Lambda, \infty).$$

In particular the endpoint has only real zeros, and for every $t < \Lambda$ the actual F_t has a nonreal zero.

This is the classical finite-threshold conclusion, not a new result. Newman's Theorem 3 [2] proves its antecedent using his measure classification. We reconstruct instead the frozen library's actual Gaussian-smoothing and elementary four-point route. In this normalization Λ is exactly the usual de Bruijn–Newman constant: Rodgers–Tao [4] use precisely our kernel, time and Xi normalization in their equations (1)–(4). Their separate Theorem 1 proves $\Lambda \ge 0$. That deep external result is credited here; its proof is not reconstructed, formalized anew or used below.

Upper-ray propagation and endpoint closedness

All the analytic hypotheses of Sections 2–4 hold with any fixed F_a in place of F_0. If every zero of F_a has $|\operatorname{Im} z| \le b$, its actual divisor polynomials share that same strip and common Gaussian budget. Their finite polynomial heat contraction, actual Gaussian integral limit and vertical strip closure prove

$$F_{a+s}(z) = 0 \quad \Longrightarrow \quad (\operatorname{Im} z)^2 \le \max(b^2 - 2s, 0) \qquad (s \ge 0).$$

These are the same successive limits as before, now at the fixed starting time a; no semigroup on an unrestricted entire-function space is needed. Taking $b = 0$ proves that once all zeros are real, they remain real at every later time. Thus the set $\mathcal{R}$ displayed in Theorem 8.1 is an upper set, and Theorem 1.1 makes it nonempty.

It is also closed. Suppose $a_n \in \mathcal{R}$ and $a_n \to a$. For a fixed n, every divisor polynomial of F_{a_n} is real rooted. Its modulus on each line $x + iy$ is nondecreasing for $y \ge 0$, by the elementary real-root factor inequality used in Lemma 6.1. Passing through its compact limit gives

$$|F_{a_n}(x + iy_1)| \le |F_{a_n}(x + iy_2)| \qquad (0 \le y_1 \le y_2).$$

Time continuity at the two fixed spatial points passes this inequality to F_a. If $F_a(x + iy) = 0$ with $y > 0$, every point $x + iv$, $0 \le v \le y$, would be a zero. The identity theorem contradicts $F_a(0) > 0$. Conjugation excludes a zero below the real axis. Hence $a \in \mathcal{R}$. What remains is to exclude $\mathcal{R} = \mathbb{R}$ by an actual negative-time obstruction.

Proposition 8.2. *For some strictly negative t, the actual F_t has a nonreal zero. No numerical time or particular zero is specified by this argument.*

The elementary four-point inequality

For $c \geq 0$ let $\ell_c(v) = \log(1 + cv)$ on $v \geq 0$, and define

$$D_c = 2\log(1 + c/2) - \log(1 + c).$$

For every $y \geq 1$,

$$\ell_c(y) \geq \ell_c(1) + (y-1)(\ell_c(2) - \ell_c(1)) - 2(y-1)^2 D_c. \tag{15}$$

To verify it, put $u = c/(1+c)$ and $v = y - 1$. Both are nonnegative. The scalar inequality $\log(1 + w) \geq w - w^2/2$ for $w \geq 0$ follows by differentiating the difference, whose derivative is $w^2/(1+w)$. Furthermore

$$D_c = \log \frac{(1 + c/2)^2}{1 + c} \geq 1 - \frac{1 + c}{(1 + c/2)^2} = \left(\frac{c}{c+2}\right)^2, \qquad u \leq \frac{2c}{c+2}.$$

Here $\log r \geq 1 - r^{-1}$ for $r > 0$ follows from $\log x \leq x - 1$ applied to $x = r^{-1}$. Consequently $u^2 \leq 4D_c$. Now $\ell_c(y) - \ell_c(1) = \log(1 + uv)$ and $\ell_c(2) - \ell_c(1) = \log(1 + u) \leq u$. Therefore

$$\log(1 + uv) \geq uv - \tfrac{1}{2}u^2v^2 \geq v\log(1 + u) - 2v^2 D_c,$$

which proves (15).

It follows by summation that every profile $q(v) = A - Bv + \sum_j \alpha_j \log(1 + c_j v)$, with $c_j, \alpha_j \geq 0$, satisfies

$$q(y) \geq q(1) + (y-1)(q(2) - q(1)) - 2(y-1)^2(2q(1/2) - q(0) - q(1)), \qquad y \geq 1. \tag{16}$$

Constant and linear functions satisfy this with equality, so adding $A - Bv$ changes no curvature term. For a nonzero polynomial P with all roots real and $P(0) \neq 0$, its complete real-root factorization gives

$$\log|P(iC\sqrt{v})| = A_P + \frac{1}{2}\sum_{r \in \mathrm{roots}(P)} \log(1 + (C/r)^2 v).$$

All roots are counted with multiplicity; each is nonzero. Hence $A - Bv + \log|P(iC\sqrt{v})|$ satisfies (16). If an entire f in our growth class has only real zeros and $f(0) \neq 0$, Section 3 gives such real-rooted approximants with the exact disk zeros and preserved origin. At every point on the imaginary axis f is nonzero: a purely imaginary real zero must be zero itself. Thus continuity of $\log|\cdot|$ passes their inequalities to $A - Bv + \log|f(iC\sqrt{v})|$ at the five fixed values $0, 1/2, 1, 2, y$. This step needs no differentiation or infinite logarithm sum.

An actual density smoothing identity

Let $F(x) = K(|x|)$ and $Y \sim N(0, 2)$. This is a bounded, strictly positive continuous even function. For $\epsilon > 0$ put $S_\epsilon(x) = \mathbb{E}F(x + \epsilon Y)$. These are genuine positive integrals. Bounded dominated convergence gives $S_\epsilon(x) \to F(x)$ as $\epsilon \downarrow 0$. The Gaussian variance is exactly $2\epsilon^2$. Pairing the two half-lines and completing the square gives

$$S_\epsilon(x) = \frac{1}{\sqrt{\pi}\epsilon} e^{-x^2/(4\epsilon^2)} F_{-1/(4\epsilon^2)}\left(\frac{ix}{2\epsilon^2}\right). \tag{17}$$

Indeed its full-line density is $(2\sqrt{\pi}\epsilon)^{-1} e^{-(u-x)^2/(4\epsilon^2)}$. The sum at u and $-u$, on $u \geq 0$, equals

$$(\sqrt{\pi}\epsilon)^{-1} e^{-x^2/(4\epsilon^2)} e^{-u^2/(4\epsilon^2)} \cosh(ux/(2\epsilon^2)).$$

Multiplying by $K(u)$ and using $\cos(iw) = \cosh w$ proves the formula. The bounded-density Gaussian integral is absolutely integrable; the half-line cosh integral is absolutely integrable by theta decay. In particular the rightmost transform value is strictly positive real, so every logarithm used below is legitimate.

Proof of Proposition 8.2. Suppose instead that F_t has only real zeros for every $t < 0$. For each $\epsilon > 0$ take $f = F_{-1/(4\epsilon^2)}$ in the preceding polynomial-limit inequality. Taking logarithms in (17) shows that $q_\epsilon(v) = \log S_\epsilon(\sqrt{v})$ is exactly one such profile with

$$A = \log(1/(\sqrt{\pi}\epsilon)), \qquad B = 1/(4\epsilon^2), \qquad C = 1/(2\epsilon^2).$$

Consequently (16) holds for q_ϵ. Fix $y \geq 1$ and send $\epsilon \downarrow 0$. Positivity of F and continuity of the real logarithm imply convergence at all five required points. Thus the same inequality holds for $q(v) = \log K(\sqrt{v})$.

Its right side is one fixed quadratic polynomial $p(y) = ay^2 + by + c$. For $y \geq 1$, $p(y) \geq -(|a| + |b| + |c|)y^2$. Put $L = |a| + |b| + |c|$. Then $\log K(x) \geq -Lx^4$ for every $x \geq 1$, equivalently

$$e^{Lx^4} K(x) \geq 1 \qquad (x \geq 1).$$

But the actual theta estimate $K(x) \leq Ce^{9x-\pi e^{4x}/2}$ gives $e^{Lx^4}K(x) \to 0$: the exponential e^{4x} dominates the fixed polynomial $Lx^4 + 9x$. This contradiction proves that at least one of the strictly negative quadratic times has a nonreal zero. □

Proof of Theorem 8.1. Let $\tau < 0$ and a nonreal zero of F_τ be supplied by the proposition. Every $r \in \mathcal{R}$ must satisfy $r > \tau$: otherwise upper-set propagation would put τ in $\mathcal{R}$. Thus $\mathcal{R}$ is nonempty and bounded below. Its infimum Λ is a finite real number. Choose $r_n \in \mathcal{R}$ with $\Lambda \leq r_n < \Lambda + 1/n$ for $n \geq 1$, using the defining infimum property. Closedness gives $\Lambda \in \mathcal{R}$; upper propagation then proves $\mathcal{R} = [\Lambda, \infty)$. Because $1/2 \in \mathcal{R}$, one has $\Lambda \leq 1/2$. For $t < \Lambda$, nonmembership means exactly that an actual nonreal zero exists. These statements include the endpoint and make no assumption about the simplicity of its zeros. □

The named formal endpoint is
`RiemannQuadraticFiniteThreshold.real_zeros_iff`, with the finite lower-bound obstruction in that same module and the smoothing identity in `RiemannQuadraticDensityHeat`. The proof here first produces a strictly negative nonreal time; the formal route first excludes all-realness at every real time and then uses its attained threshold to produce such a negative time. This explicitly labelled human reordering proves the same conclusions.

By the exact Xi normalization, $\Lambda \leq 0$ is the standard threshold formulation of RH. This paper proves neither that bound nor the numerical value of Λ. The separate published lower bound $\Lambda \geq 0$ is not an open sign question in the literature and is not a consequence of the elementary proof above. The bound $1/2$ is retained solely as the classical bound reconstructed here, without claiming it is the best available bound. No quartic or sixth-order threshold is identified with Λ.

9 Formal correspondence, primary comparison and verification

The map supplied with this paper records exact declarations, source hashes and line links at the fixed baseline. In the source, F_t is `RiemannEvenFlow.flow1t` and H_t is `PureOrderPolynomialTransport.transport2(-t)`. The main chain is:

- `RiemannQuadraticZeroProductTails.zero_time_strip` derives width one from the actual zeta nonvanishing theorem for $\operatorname{Re} s > 1$, without an external 7/8 premise.
- `RiemannQuadraticPolynomialHeat` supplies the actual even polynomials, disk-zero retention, normalization and common Gaussian budget.
- `RiemannQuadraticUniformHeat` and `RiemannQuadraticOperatorApproximation` identify genuine locally uniform Gaussian and polynomial-operator limits.
- `SymmetricPolynomialStrip`, `SymmetricPolynomialTaylor` and `SymmetricPolynomialHeatLimit` retain multiplicities, the scaled shift budget and the fixed-degree heat limit.
- `PolynomialStripLimit` proves strip closure and finite polynomial heat contraction.
- `RiemannQuadraticGlobalStrip.flow_zero_strip` and `real_zeros_after_half` give exactly Theorem 1.1.

The proof above uses classical Hadamard factorization with Jensen's estimate for the divisor-polynomial construction, and a finite-dimensional coefficient telescoping proof for the Euler step. These are explicit human presentations of the named formal conclusions, rather than shorter new Lean declarations. The source's annular divisor-tail and factorial-jet proofs remain separately mapped. Fundamental complex-analysis inputs (identity theorem, Jensen and Hadamard), the elementary theta identity, Mellin continuation and the Euler-product zero-free half-plane are stated where used. No new Lean proof or research-source modification is made.

The classical status of the main bound is settled by the exact primary antecedent: Newman's 1976 Theorem 1 states the broader strip contraction and credits de Bruijn. Its complete statement was inspected, including its decay and evenness hypotheses and its growing quadratic multiplier. Its strict initial strip can be widened from width one to $1 + \epsilon$ and then $\epsilon \downarrow 0$, giving the closed-boundary convention used here. The original de Bruijn full-text comparison remains incomplete in this editorial audit; the source's citation identifies its finite-shift route, but that is not presented as a fresh full-text review by the editor. McMullen's primary course Hadamard passages provide the foundational factorization context. No novelty is claimed for any general strip, Gaussian heat or entire-product mechanism. The contribution of this companion is a readable, exact-domain reconstruction of an actual verified arithmetic proof chain. A reviewed draft is not an external submission.

The frozen baseline already has a complete successful Lean root build and transitive axiom audit, with the standard classical axioms only. Publication work checks those recorded evidence hashes and all mapped source bytes; it does not consume a redundant research build. Independent finite algebra checks, when supplied, verify only the displayed pair identity and finite heat/shift transcriptions, not the analytic theorems. The source/PDF compilation and visual-review receipts are kept separately.

References

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[3] C. McMullen, *Complex Analysis*, Harvard Math 213a course notes, 2023, Theorems 3.15 and 3.18, printed pp. 80–82. https://people.math.harvard.edu/~ctm/home/text/class/harvard/213a/23/html/home/course/course.pdf.

[4] B. Rodgers and T. Tao, *The de Bruijn–Newman constant is non-negative*, author manuscript v5, July 3, 2021, equations (1)–(4) and Theorem 1, printed pp. 2–3. https://arxiv.org/abs/1801.05914. The external nonnegativity proof is not reproduced here.